CRC Lab — Viva Preparation Sheet

Read this once, then walk in. Total demo: about 2 minutes.

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1. The one paragraph (memorize this)

CRC appends $n-1$ zeros to the data, divides by the generator using XOR, and the leftover is the CRC. We transmit data + CRC. The receiver divides what arrives by the same generator: remainder zero means no error, non-zero means a bit flipped. XOR replaces subtraction because binary arithmetic has no borrows. If they ask nothing else, say this and offer to type their example live.

2. Minute-by-minute demo

1. **Inputs (30s).** Default `HELLO` + `1011`. Read the numbers live: data bits, 3 appended zeros, remainder, frame.
2. **Division (30s).** Press Play. Narrate: quotient on top, one staggered subtraction row per step, $\oplus$ where the leading 1 was cancelled.
3. **Error (30s).** Flip exactly one bit. Red bit, remainder non-zero, verdict ERROR. Say: one flipped bit, caught.
4. **Reset + live input (30s).** Reset to VALID. Ask the examiner for a word, type it. Everything recomputes.
5. **Report (15s).** Hit Export, show the PDF matches the screen: same inputs, every division step, same verdict.

Golden rule: Reset before every flip demo. One flip, show, reset. Never stack many flips on stage (Section 4 explains why).

3. Likely questions, one-line answers

- **Why append $n-1$ zeros?** To make room for the remainder; a remainder modulo an n-bit generator is always $n-1$ bits.
- **Why must the generator start and end with 1?** Leading 1 sets its degree; trailing 1 means it catches all single-bit errors.
- **Why XOR, not subtraction?** In mod-2 arithmetic they are identical; XOR just has no borrows.
- **Does CRC correct errors?** No, detection only. It says *that* something changed, never what the original was.
- **Text vs Bits mode?** Text converts each character to 8-bit ASCII first; the division always runs on bits.
- **What did you build vs reuse?** Everything: the division engine, both UIs, the LaTeX exporter. Only `pdflatex` itself is external.

4. The trap: VALID with flipped bits

If you flip several bits, their remainders can XOR to zero and cancel, so the verdict reads VALID. This is genuine theory, verified during development: single-bit errors are always caught; some multi-bit patterns are exact multiples of the generator and slip through. If the examiner spots it, say exactly that. Then reset and demo a single flip.

5. Day checklist

- `python3 report/server.py` running before they arrive; browser open at `http://127.0.0.1:8321/`.
- TUI binary tested once in their terminal size (`./target/release/crc-tui`): `d` edits message, `g` edits generator, `space` plays, `e` flips, `x` resets, `t` changes theme, `?` help.
- One sample PDF in hand or on disk.
- If stuck on any question: return to the memorized paragraph and offer a live example.